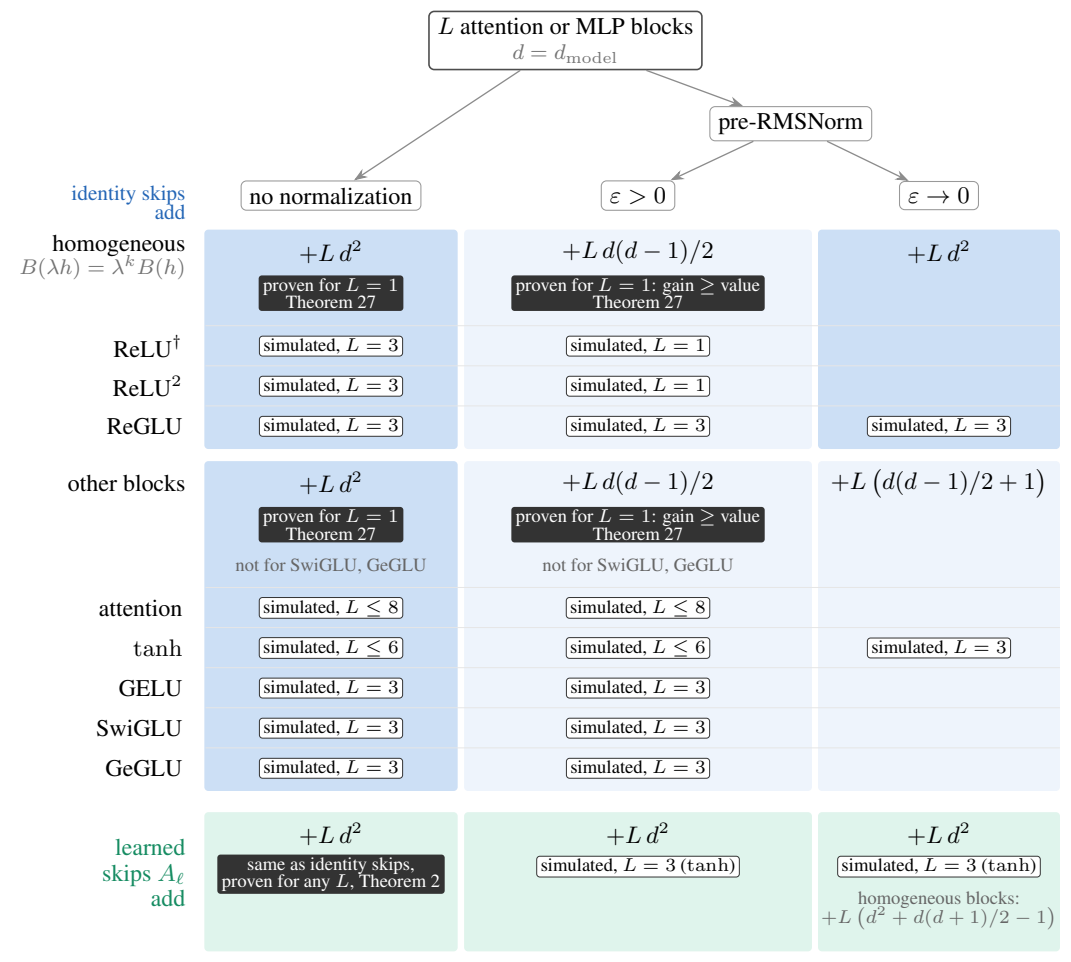

L attention or MLP blocks
$d = d_{\text{model}}$
pre-RMSNorm
identity skips add
no normalization
$\varepsilon > 0$
$\varepsilon \to 0$
homogeneous $B(\lambda h) = \lambda^k B(h)$
$+L\,d^2$
proven for $L = 1$ Theorem 27
$+L\,d(d-1)/2$
proven for $L = 1$: gain $\geq$ value Theorem 27
$+L\,d^2$
ReLU†
simulated, $L = 3$
simulated, $L = 1$
ReLU^2
simulated, $L = 3$
simulated, $L = 1$
ReGLU
simulated, $L = 3$
simulated, $L = 3$
simulated, $L = 3$
other blocks
$+L\,d^2$
proven for $L = 1$ Theorem 27
not for SwiGLU, GeGLU
$+L\,d(d-1)/2$
proven for $L = 1$: gain $\geq$ value Theorem 27
not for SwiGLU, GeGLU
$+L\left(d(d-1)/2+1\right)$
attention
simulated, $L \leq 8$
simulated, $L \leq 8$
tanh
simulated, $L \leq 6$
simulated, $L \leq 6$
simulated, $L = 3$
GELU
simulated, $L = 3$
simulated, $L = 3$
SwiGLU
simulated, $L = 3$
simulated, $L = 3$
GeGLU
simulated, $L = 3$
simulated, $L = 3$
learned skips A_ℓ add
$+L\,d^2$
same as identity skips, proven for any L, Theorem 2
$+L\,d^2$
simulated, $L = 3$ (tanh)
$+L\,d^2$
simulated, $L = 3$ (tanh)
homogeneous blocks: $+L\left(d^2 + d(d+1)/2 - 1\right)$